

Mathematical Physics I (Fall 2025): Homework #6

Due Dec. 5, 2025 (Fri, 23:00pm)

[0.5 pt each, total 5 pts, turn in as a single pdf file to eTL]

- By turning in your homework, you acknowledge that you have not received any unpermitted aid, nor have compromised your academic integrity during its preparation. (Remember the SNU College of Natural Sciences Honor Code!)
- Textbook problem numbering convention (Boas, 3rd ed.): [Section].[Problem No.]. Many problems in this assignment are from the list of suggested problems announced in the class.
- Exhibit all intermediate steps to receive full credits. Only handwritten answers are accepted except for numerical problems or verifications — for which you print out and turn in not just the end results (e.g., plots) but also the source codes in your favorite tools such as MATLAB or MATHEMATICA.

1.-3. Boas Chapter 8, Problems 11.9, 11.15, 12.4

(Note: For Problem 11.9, you will first want to review Problem 11.6(b) and (c), and Example 10.1. You may assume $y_0 = y'_0 = 0$ as in Problem 11.6(a). You may opt to utilize the *translation theorems* of the Laplace transform described in Problems 8.19 to 8.27 of Boas Chapter 8. For Problem 12.4, note that we assume $t > 0$ and $y_0 = y'_0 = 0$ at $t = 0$; see the text right before Eq.(12.6) in Boas Chapter 8. Compare your answer with what is given in Problem 10.18. Sketch the motion as discussed in Problem 10.18. To refresh your memory about the Green function of a harmonic oscillator, you may also wish to briefly review the classical mechanics textbooks such as Thornton & Marion (Chapter 3.9).)

4.-6. Boas Chapter 13, Problems 2.1, 3.7, 4.5

(Note: For Problem 2.1 to 4.5, carefully read the instruction at the beginning of the Problems; that is, you are asked to make a computer plot using the first several terms of your answer. For Problem 3.7, you may want to review the discussion about the *Insulated boundaries* in Boas Chapter 13, Section 3.)

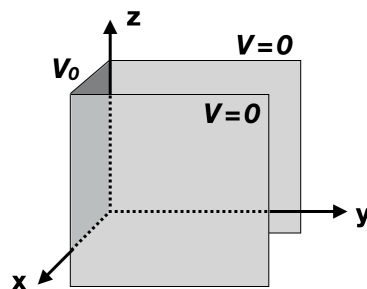
7.-8. Boas Chapter 9, Problems 2.10, 4.4

(Note: For Problem 4.4, as you evaluate the integral for T you may utilize the indefinite integral $\int \frac{dx}{\sqrt{ax^2+bx+c}} = -\frac{1}{\sqrt{-a}} \sin^{-1}\left(\frac{2ax+b}{\sqrt{b^2-4ac}}\right)$ (if $a < 0$ and $b^2 > 4ac$) found in common integral tables like Appendix E of Thornton & Marion, or in extensive references like Zwillinger (Section 5.4.13, 33rd ed.).)

9. In the class we discussed the resemblance between Laplace's equation which describes the steady-state temperature in a region with no heat source, and the same equation as it appears in electrostatics where it describes the electrostatic potential in a charge-free region. The latter was left as an exercise for you.

(a) Review how Laplace's equation $\nabla^2 V = 0$ is acquired from Coulomb's law in Boas Chapter 6, Section 10, where V is the electrostatic potential in a charge-free region.

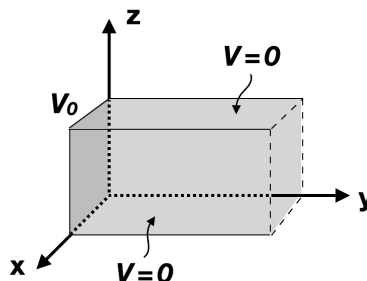
(b) Two infinite grounded metal plates lie parallel to the y - z plane, one located at $x = 0$ and the other at $x = l$. The region of our interest is the space (the "slot") between these plates. At the left boundary ($y = 0$), the slot is closed off by an infinite strip held at a constant potential V_0 , insulated from both plates. Far from the strip ($y \rightarrow \infty$), the potential approaches zero (as it should). Determine the potential $V(x, y)$ inside this "slot" for $y > 0$.



(c) Repeat the exercise in (b), but now assume that the boundary at $y = 0$ consists of two metal strips: one held at a constant potential V_0 in $x = [0, \frac{l}{2}]$, and the other held at $-V_0$ in $x = [\frac{l}{2}, l]$. Make a computer plot using the first several terms of your answer.

(d) We now extend the problem in (b) to higher dimensions. Consider an infinitely long rectangular metal "pipe" with cross-sectional sides l_x and l_z , along the x - and z -axis, respectively, which is grounded. The region of our interest is the hollow interior of the pipe. At the left boundary ($y = 0$), the pipe is closed off by a rectangular plate held at a constant potential V_0 , insulated from the pipe walls. Determine the potential $V(x, y, z)$ inside this "pipe" for $y > 0$ and demonstrate how your answer in (b) changes in this genuine 3-dimensional problem. You are asked to explicitly separate the variables in $\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$, by assuming a solution of the form $V(x, y, z) = X(x)Y(y)Z(z)$, to reach

$$V(x, y, z) = \frac{16V_0}{\pi^2} \sum_{\text{odd } n_x, n_z} \frac{1}{n_x n_z} e^{-\pi y \sqrt{\left(\frac{n_x}{l_x}\right)^2 + \left(\frac{n_z}{l_z}\right)^2}} \sin \frac{n_x \pi x}{l_x} \sin \frac{n_z \pi z}{l_z}.$$



(Note: The Laplace's equation discussed here should look familiar to most of you as you have studied this basic set up in your electromagnetism classes. If not, you may want to briefly review the electromagnetism textbooks such as Griffiths.)

10. In the class we discussed the resemblance between the heat flow equation and the Schrödinger equation. The latter was left as an exercise for you.

(a) Review how the *time-independent* Schrödinger equation, Eq.(3.22), and the time equation, Eq.(3.21), are acquired from the Schrödinger equation, Eq.(1.6), in Boas Chapter 13.

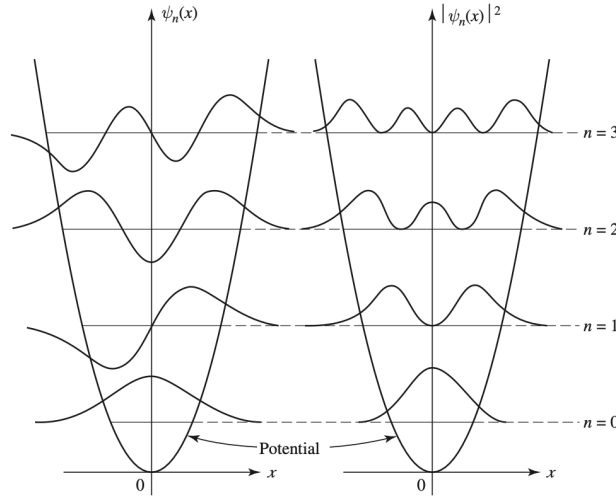
(b) Then, review the 1-dimensional “particle in a box” problem in Example 3 of Boas Chapter 13, Section 3. That is, starting from Eq.(3.23) with $V = 0$ on $[0, l]$ and boundary conditions $\Psi(0, t) = 0$ and $\Psi(l, t) = 0$, show how you can reach $\Psi(x, t)$ in Eq.(3.26) with $E_n = \frac{\hbar^2}{2m} \left(\frac{n\pi}{l}\right)^2$.

(c) Now we consider an initial condition. Verify explicitly that an initial condition $\Psi(x, 0) = \frac{100}{l}x$ yields $\Psi(x, t)$ in Eq.(3.27). Repeat the exercise and find $\Psi(x, t)$ with a different initial condition $\Psi(x, 0) = 1$ on $[0, l]$.

(d) We now extend the problem in (b) to higher dimensions. Show how your answer in (b) changes if you tackle the 3-dimensional “particle in a box” problem (e.g., a box with planar faces at $x = 0, x = l_x, y = 0, y = l_y, z = 0, z = l_z$). You are asked to explicitly separate the variables in Eq.(3.22) with $V = 0$, by assuming a solution of the form $\psi(x, y, z) = X(x)Y(y)Z(z)$, to reach

$$\Psi(x, y, z, t) = \sum_{n_x, n_y} A_{n_x n_y} \sin \frac{n_x \pi x}{l_x} \sin \frac{n_y \pi y}{l_y} \sin \frac{n_z \pi z}{l_z} e^{-iEt/\hbar},$$

where n_z is determined by $E = \frac{\hbar^2 \pi^2}{2m} \left[\left(\frac{n_x}{l_x}\right)^2 + \left(\frac{n_y}{l_y}\right)^2 + \left(\frac{n_z}{l_z}\right)^2 \right]$ for a given n_x, n_y , and E .



[Adapted from Griffiths & Schroeter, Chapter 2.3]

(e) Finally, we extend the problem in (b) to the case with a harmonic potential — i.e., a particle in a potential field $V = \frac{1}{2}kx^2$. Starting from the equation,

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2\psi = E\psi,$$

(i.e., Eq.(3.23) of Boas Chapter 13 with $V = \frac{1}{2}kx^2$, or Eq.(2.45) of Griffiths & Schroeter) and changing the variable with $\xi = \alpha x = \left(\frac{mk}{\hbar^2}\right)^{\frac{1}{4}} \cdot x$ and $\lambda = \frac{2E}{\hbar} \left(\frac{m}{k}\right)^{\frac{1}{2}}$, demonstrate that $\psi(\xi)$ satisfies the equation,

$$-\frac{d^2\psi}{d\xi^2} + \xi^2\psi = \lambda\psi.$$

Then, assuming a solution of the form $\psi(\xi) = y(\xi)e^{-\xi^2/2}$, verify that $y(\xi)$ satisfies

$$y'' - 2\xi y' + (\lambda - 1)y = 0,$$

which is the Hermite equation seen in Eq.(22.14) of Boas Chapter 12, Section 22 (with $\lambda = 2n + 1$), or in Eq.(2.79) of Griffiths & Schroeter. From this, prove that the possible wave functions for the harmonic oscillator potential are written as

$$\psi_n(\xi) = A_n H_n(\xi) e^{-\xi^2/2},$$

where $H_n(\xi)$ is the n th Hermite polynomial with the corresponding energy $E_n = (n + \frac{1}{2}) \hbar (\frac{k}{m})^{\frac{1}{2}}$.

(Note: The Schrödinger equation discussed here should look familiar to most of you as you have studied the basic “particle in a box” problem in your elementary physics classes. If not, you may want to briefly review the freshman physics textbooks such as Halliday & Resnick, or the quantum mechanics textbooks such as Griffiths & Schroeter. For (e), briefly discuss how one could come up with the educated guess $\psi(\xi) = y(\xi)e^{-\xi^2/2}$. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library’s proxy service, see <http://library.snu.ac.kr/using/proxy>.)